

7. Integration Techniques

7.3. Trigonometric Substitutions

7.3a Learning Objectives

- I can understand the relationship between right triangles and substitutions.
- I can draw triangles to represent substitutions.
- I can understand when and how to use trigonometric substitutions.

Suppose we want to derive the formula for area of a circle of radius r . We know how to set up the integral for this using area between curves:

$$\begin{aligned} A &= \int_{-r}^r \underbrace{\sqrt{r^2 - x^2}}_{\text{top}} - \underbrace{\left(-\sqrt{r^2 - x^2}\right)}_{\text{bottom}} dx \\ &= 2 \int_{-r}^r \sqrt{r^2 - x^2} dx \end{aligned}$$

Up until now, we don't have a method to evaluate this integral. Looking at a point on the circle we can use trig functions to rewrite the expressions in x and y into r and θ . We have a few options, so let's do both and compare.

Cosine	Sine
$x = r \cos(\theta)$	$x = r \sin(\theta)$
$dx = -r \sin(\theta) d\theta$	$dx = r \cos(\theta) d\theta$

could write $\cos(\theta) = \frac{x}{r}$ or $x = r \cos(\theta)$. This gives us

Find the area of a circle with radius r using techniques of integration.

7.3a Section Summary: